

SHORT COMMUNICATION

Drug resistance pattern among ART-naïve clients attending an HIV testing and counseling center in Dhaka, Bangladesh

Sezanur Rahman¹  | Md S. Sarker¹ | Shambhu G. Aralaguppe² | Golam Sarwar³ | Sharful I. Khan³ | Mustafizur Rahman¹ 

¹Virology Laboratory, Infectious Diseases Division, International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), Dhaka, Bangladesh

²Division of Clinical Microbiology, Department of Laboratory Medicine, Karolinska Institutet, Stockholm, Sweden

³HIV Programme, Infectious Diseases Division, International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), Dhaka, Bangladesh

Corresponding

Mustafizur Rahman, Virology Laboratory, Infectious Diseases Division, International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), Dhaka 1212, Bangladesh.
Email: mustafizur@icddr.org

Funding information

Swedish Research Links 2015 (Development Research) - International Collaboration, Grant/Award Number: 2015-05914

Abstract

In Bangladesh, antiretroviral therapy (ART) is provided without screening drug resistance-associated mutations (DRM) among people living with HIV, while DRM might emerge and transmit to the newly infected individual. The present study was aimed to identify DRM among ART-naïve clients from an HIV testing and counseling (HTC) center in the initial stages of ART programs. Randomly selected ($n = 64$) archived plasma samples were used for the *pol* gene amplification and sequencing by sanger technology. Recovered sequences ($n = 10$) were genotyped using HIV genotyping tools of NCBI and analyzed using the Stanford University HIV drug resistance database (hivdb.stanford.edu). Various genotypes with a number of DRM were identified in HTC clients, who belonged to different risk groups based on behavioral data. The drug resistance algorithm showed that all samples were fully resistant to tipranavir/ritonavir drugs except for one intermediate resistance. Despite the small sample size, our understanding from this study warrants an ART policy with a DRM monitoring system for the country.

KEYWORDS

ART-naïve, Bangladesh, drug resistance-associated mutations, HIV

1 | INTRODUCTION

Given the low HIV prevalence among general populations in Bangladesh ($<0.01\%$), National HIV serological surveillance remained focused on key populations (KPs) at risk of HIV since 1998, where the average HIV prevalence was found as 3.9% .¹ HIV-1 genotyping data indicated a prevalence of subtype C and C-related recombinants among people who inject drugs (PWID), which might have originated from neighboring countries.^{2,3} Other HIV-1 strains in the general population appeared to have been introduced by returnee migrants from the Middle East and transmitted through the sexual routes.^{2,4,5}

Antiretroviral (ARV) drugs/antiretroviral therapy (ART) is considered the most effective measure to improve the survival and well-being of patients who are living with HIV.^{6,7} In 2005, ART was started in Bangladesh by the NGOs, later this program was expanded through Global Fund. Since October 2017, AIDS/STD Program had

taken over complete treatment care services⁸; the success of the ART program is contingent on the emergence of drug-resistant HIV strains. At the population level, drug resistance detection can warn program planners about the existence and circulation of drug-resistant viral mutations. At the individual patient level, to optimize therapeutic regimens, treatment guidelines recommend regular testing to detect drug resistance-associated mutations (DRM) in patients before they initiate ART.^{9,10} Though HIV viral load monitoring is now available on a limited scale in Bangladesh, HIV drug resistance testing is not available yet for ART centers. Limited data from the latest HIV sero-surveillance conducted in 2016 indicated the presence of DRM among PWID.³

In this context, in 2016, the Virology Laboratory of International Centre for Diarrheal Disease Research, Bangladesh (icddr,b), and Department of Laboratory Medicine, Karolinska Institutet, Sweden, started a collaborative program to transfer the technology for drug

resistance testing by applying HIV-1 genome sequencing. Through this collaboration, we conducted the present study to identify drug resistance mutations among ART-naïve clients from an HIV testing and counseling (HTC) center in the initial stages of ART programs.

2 | MATERIALS AND METHODS

In Bangladesh, the first HTC unit was “Jagori,” established by the icddr,b in January 2002 to provide HTC services in a confidential environment to populations at risk for HIV, including the general population until 2015.¹¹ Informed written consents were obtained from all the clients using a consent form, specifying that the client's specimen might be used for future research and that their informational confidentiality would be ensured. All patient records and information were anonymized and deidentified before analysis.

A total of 64 stored (−80°C) plasma samples were randomly selected, and viral RNA was extracted using the QIAamp Viral RNA Mini Kit according to the manufacturer's protocol (QIAGEN). A nested polymerase chain reaction (PCR) amplified the entire *pol* gene after one-step RT-PCR. SuperScript™ III, One-Step RT-PCR System with Platinum™ Taq DNA Polymerase (Invitrogen), was used in first-round PCR, and KAPA HiFi HotStart ReadyMix (KapaBiosystems) was used in the second-round PCR using gene-specific primers.¹² The amplified products were purified and sequenced in Applied Biosystems® 3500 Genetic Analyzer. The chromatograms of the sequences were manually analyzed for errors and assembled using Lasergene applications (DNASTAR). All sequences generated in this study were submitted to the GenBank database under accession no. MW270148 – MW270168.

BLAST (blast.ncbi.nlm.nih.gov) tool was used to identify the most similar global strains. Genotyping was also performed using HIV genotyping tools of NCBI (ncbi.nlm.nih.gov/projects/genotyping/formpage.cgi) using the 2009 reference set. Drug resistance profiles and amino acid substitution sites were obtained from the Stanford University HIV drug resistance database (Program Version 2.2.11, Algorithm Version 8.8; hivdb.stanford.edu/). Concurrently, the quality of the *pol* gene was checked using the HIVseq programs. The HIValg program was used to compare drug resistance profiles with Rega Institute (KU Leuven; v10.0.0) and Agence Nationale de Recherches sur le SIDA (ANRS; v29).

3 | RESULTS

We recovered nine sequences for the protease (*prot*) region and ten for the reverse transcriptase (RT) and integrase (*int*) regions. The recovered samples were from individuals aged 18–40-year old (Table 1), of whom eight were male, and the remaining two were female. Risk behavior data showed that four samples were obtained from PWID, two from general populations and one each from the migrant worker, TB patient, a client of the sex worker, and groups carrying multiple risks. Genotyping results showed high diversity among KPs with B/C recombinant ($n = 4$), CRF01/B recombinant ($n = 2$), CRF49_cpx ($n = 2$), CRF30_0206 ($n = 1$), and CRF43_02G ($n = 1$).

Amino acid substitution chart (Figure 1) showed that all three genes (*prot*, *RT*, and *int*) have major and/or accessory DRM sites, which might lead to drug resistance. Upon drug resistance analysis, two algorithms (HIVdb and REGA) showed that all samples were susceptible to protease inhibitors (PIs), nucleoside/nucleotide reverse

TABLE 1 Risk behavior, genotype, and drug-resistant profile (ANRS drug resistance algorithm) of the study strains

Lab ID	Group/risk behavior	Age (years)	Sex	Data collected	Pol genotype	ATV/r PI ^a	SQV/r	TPV/r	BIC INSTI ^a	DTG	EVG	RAL
BD01	External migrant worker	40	M	23.06.13	CRF59_01B	S	S	I	S	S	S	S
BD02	General population	34	M	02.06.13	CRF49_cpx	–	–	–	S	S	S	S
BD10	Client had a TB infection	40	M	04.11.15	CRF08_BC	S	S	R	S	S	S	S
BD24	PWID	39	F	02.04.12	CRF08_BC	S	S	R	I	I	R	R
BD30	Client of sex worker	35	M	21.05.12	CRF49_cpx	I	S	R	S	S	S	S
BD31	Multiple	25	M	21.05.12	CRF43_02G	S	I	R	S	S	S	S
BD34	PWID	18	M	03.02.13	CRF34_01B	S	S	R	S	S	S	S
BD36	PWID	35	M	04.08.13	CRF08_BC	S	S	R	I	I	R	R
BD53	General population	21	F	26.12.13	CRF30_0206	S	S	R	S	S	S	S
BD55	PWID	32	M	03.02.14	CRF57_BC	S	S	R	S	S	S	S

Note: The drugs that showed susceptibility to all samples are not presented in the table.

Abbreviations: ATV/r, atazanavir/ritonavir; BIC, bictegravir; DTG, dolutegravir; EVG, elvitegravir; INSTI, integrase strand transfer inhibitor; PI, protease inhibitor; PWID, people who inject drug; RAL, raltegravir; SQV/r, saquinavir/ritonavir; TPV/r, tipranavir/ritonavir.

^aBased on sequence data analyzed on the ANRS drug resistance algorithm. S, susceptible; I, intermediate resistant; R, resistant; –, sequence data not available.

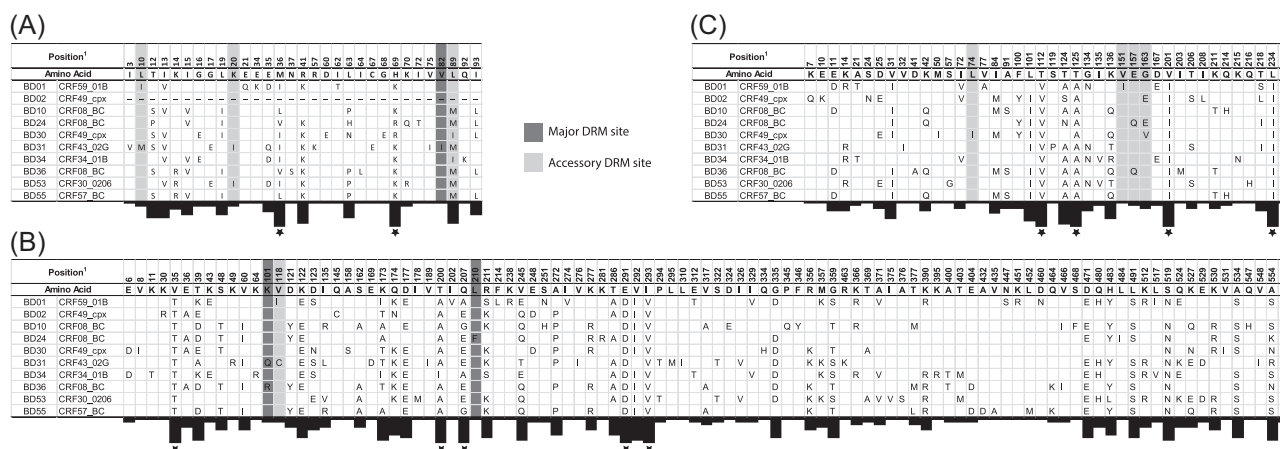


FIGURE 1 Amino acid substitution chart for (A) protease, (B) reverse transcriptase, and (C) integrase gene of the study strains. The amino acid position was based on HXB2 numbering, and letters represent the standard one-letter amino acid code. Shading indicates the drug resistance-associated mutations site that presents in the study strains

transcriptase inhibitors (NRTIs), nonnucleoside reverse transcriptase inhibitors (NNRTIs), and integrase strand transfer inhibitor (INSTI) drugs. However, the ANRS drug resistance algorithm showed that all the samples were susceptible to NNRTI and INSTI but fully resistant to PIs drug tipranavir/ritonavir (TPV/r) except for one intermediate resistant (Table 1). Two samples showed complete resistance to INSTI drug elvitegravir, raltegravir, and intermediate resistance to dolutegravir and bictegravir. Also, one sample was intermediate resistant to atazanavir/ritonavir and another to saquinavir/ritonavir (Table 1).

4 | DISCUSSION

This study identified diverse circulating genotypes and the presence of DRM among ART-naïve HTC clients. In the HTC, different KPs at risk of HIV had come to know their HIV status. These clients were classified into several risk groups based on behavioral data. A substantial proportion of the clients were migrant workers and their spouses and children, followed by PWID, sex worker, a client of the sex worker, and TB patient.^{4,11} Due to different infection sources in diverse risk groups, high HIV-1 genotypic diversity was anticipated among HTC clients, which was confirmed by identifying five different genotypes in 10 recovered clients. However, the samples were preserved for a prolonged time, which might have undergone multiple freeze-thaw cycles while being used for other diagnostics/studies, resulting in a low success rate (16%) of sequence recovery.

Our data provide evidence that DRM was present among newly diagnosed ART-naïve patients from the early onset of the ART program. The possible explanation is that the DRM could be positively selected for the transmitted variants¹³ and later circulated within the population¹⁴; therefore, TPV/r drug resistance identified in our study also persisted.³ Despite the small sample size, our understanding from this study warrants an established ART policy with a monitoring system for resistance-associated mutations.

ACKNOWLEDGMENTS

The study was conducted through Swedish Research Council Grant. icddr,b acknowledges with gratitude the commitment of the Swedish Research Council and Karolinska Institutet to its research efforts. icddr,b is also grateful to the Government of Bangladesh, Canada, Sweden, and the United Kingdom for providing core/unrestricted support. The authors also acknowledge Dr. Ujjwal Neogi from the Division of Clinical Microbiology, Department of Laboratory Medicine, Karolinska Institutet, for providing intellectual support.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

Mustafizur Rahman and Md S. Sarker conceived and coordinated the study and Sezanur Rahman wrote the paper. Sezanur Rahman and Shambhu G. Aralaguppe performed laboratory procedures, including nucleic acid extraction, PCR, sequencing, and genotyping. Sezanur Rahman and Shambhu G. Aralaguppe performed data analysis and interpretation of the results. Mustafizur Rahman, Md S. Sarker, Golam Sarwar, and Sharful I. Khan reviewed the article critically for intellectual content. All authors reviewed subsequent drafts of the manuscript and approved the final version.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the manuscript.

ORCID

Sezanur Rahman <http://orcid.org/0000-0002-0092-9261>

Mustafizur Rahman <http://orcid.org/0000-0002-6876-0191>

REFERENCES

1. National AIDS/STD Programme 2017. *Behavioural and Serological Surveillance amongst Key Populations at Risk of HIV in Selected Areas*

- of Bangladesh, 2016. Government of the People's Republic of Bangladesh: Ministry of Health and Family Welfare, Welfare MoHaF; 2017.
2. Bontell I, Sarker MS, Rahman M, Afrad MH, Sonnerborg A, Azim T. Molecular dating of HIV-1 subtype C from Bangladesh. *PLoS One*. 2013;8(11):e79193.
 3. Rahman M, Rahman S, Reza MM, Khan SI, Sarker MS. HIV-1 drug resistance and genotypes circulating among HIV positive key populations in Bangladesh: 2016 update. *Int J Infect Dis*. 2020;104:150-158.
 4. Shahrin L, Leung DT, Matin N, Kawser CA, Pervez MM, Chisti MJ. Clinical profile of hospitalized HIV-infected children in Bangladesh, a low-HIV-prevalence country. *Paediatr Int Child Health*. 2014;34(2):133-137.
 5. Sarker MS, Azim T, Islam LN, Rahman M. Migrant workers play a key role in HIV-1 strain diversity in Bangladesh. *HIV & AIDS Rev*. 2020;19(4):278-283.
 6. Hammer SM, Saag MS, Schechter M, et al. Treatment for adult HIV infection: 2006 recommendations of the International AIDS Society-USA panel. *JAMA*. 2006;296(7):827-843.
 7. Antiretroviral Therapy Cohort Collaboration. Survival of HIV-positive patients starting antiretroviral therapy between 1996 and 2013: a collaborative analysis of cohort studies. *Lancet HIV*. 2017;4(8):e349-e356.
 8. National AIDS/STD Programme. *National Anti Retroviral Therapy (NRT) Guidelines, Bangladesh*. Government of the People's Republic of Bangladesh: Directorate General of Health Services; 2019.
 9. Vandamme AM, Camacho RJ, Ceccherini-Silberstein F, et al. European recommendations for the clinical use of HIV drug resistance testing: 2011 update. *AIDS Rev*. 2011;13(2):77-108.
 10. Thompson MA, Aberg JA, Hoy JF, et al. Antiretroviral treatment of adult HIV infection: 2012 recommendations of the International Antiviral Society-USA panel. *JAMA*. 2012;308(4):387-402.
 11. Urmi AZ, Leung DT, Wilkinson V, Miah MA, Rahman M, Azim T. Profile of an HIV testing and counseling unit in Bangladesh: majority of new diagnoses among returning migrant workers and spouses. *PLoS One*. 2015;10(10):e0141483.
 12. Grossmann S, Nowak P, Neogi U. Subtype-independent near full-length HIV-1 genome sequencing and assembly to be used in large molecular epidemiological studies and clinical management. *J Int AIDS Soc*. 2015;18:20035.
 13. Little SJ, Frost SD, Wong JK, et al. Persistence of transmitted drug resistance among subjects with primary human immunodeficiency virus infection. *J Virol*. 2008;82(11):5510-5518.
 14. Ross LL, Shortino D, Shaefer MS. Changes from 2000 to 2009 in the prevalence of HIV-1 containing drug resistance-associated mutations from antiretroviral therapy-naïve, HIV-1-infected patients in the United States. *AIDS Res Hum Retroviruses*. 2018;34(8):672-679.

How to cite this article: Rahman S, Sarker MS, Aralaguppe SG, Sarwar G, Khan SI, Rahman M. Drug resistance pattern among ART-naïve clients attending an HIV testing and counseling center in Dhaka, Bangladesh. *J Med Virol*. 2022;94:787-790. doi:10.1002/jmv.27387